

Global E-Commerce Sales Analysis

Excel + SQL + Python Data Analysis Project

Project Overview

An end-to-end e-commerce data analysis project using **Excel, SQL, and Python** to clean, transform, analyze, and visualize sales data and identify key business insights.

Dataset Overview

Metric	Value
Orders	2,000
Unique Customers	1,534
Total Quantity Sold	7,115 units
Columns	15
Period	2023–2025
Duplicate Rows	0

Key Business KPIs

Total Sales	484,559
Total Profit	158,872
Total Orders	2,000
Customers	1,534
Quantity	7,115

Excel Analysis

- Cleaned and prepared e-commerce data for analysis.
- Used Pivot Tables, Pivot Charts, KPI cards, and trend analysis.
- Europe generated the highest regional sales, followed by North America and Asia Pacific.
- Analyzed sales, profit, quantity, pricing, discounts, shipping cost, regions, countries, categories, customer segments, and payment methods.

SQL Analysis

- Used SQL for filtering, grouping, aggregation, and comparative analysis.
- Analyzed sales and customer data to identify trends, product performance, regional performance, and business metrics.
- SQL analysis was combined with Excel and Python findings for an end-to-end analytical workflow.

Python Analysis

- Loaded and analyzed the dataset using Pandas.
- Checked shape, columns, data types, missing values, and duplicates.

- Converted Order_Date into datetime format.
- Calculated total sales, profit, orders, customers, and quantity.
- Performed groupby analysis by region, category, segment, country, product, and payment method.
- Created monthly sales/profit trends, top-product charts, and correlation analysis using Python visualization libraries.

Key Insights

- Total sales reached 484,559 with total profit of 158,872.
- Europe generated the highest regional sales at 137,006.
- North America generated 133,876 and Asia Pacific generated 121,708.
- The dataset contains 2,000 orders and 1,534 unique customers.
- Product, customer-segment, regional, and payment-method analysis provides multiple business decision views.
- Monthly trend analysis helps identify changes in sales and profit over time.

Tools & Skills

Excel: Data Cleaning, Pivot Tables, Pivot Charts, KPI Analysis

SQL: Filtering, GROUP BY, Aggregation, Comparative Analysis

Python: Pandas, NumPy, Matplotlib, Seaborn

Core Skills: Data Cleaning, EDA, Data Analysis, Aggregation, Visualization, Business Insights

Project Purpose

This project demonstrates practical data analyst skills by combining spreadsheet analysis, SQL querying, and Python-based analysis and visualization in a single business-focused workflow.